Armin Bazarjani

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Education

University of Southern California — Los Angeles, California

Ph.D., Computational Neuroscience — 2023 – 2028 (expected)

Research focus: Computational and cognitive neuroscience and machine learning — reinforcement learning, representation learning, and world models.

Master of Science, Electrical Engineering (3.8 GPA) — 2020 – 2021

Bachelor of Science, Electrical Engineering — 2016 – 2020

Publications

Armin Bazarjani & Payam Piray "Default Feature Representations of the Cognitive Map" (2026) *NeurIPS* PDF

Armin Bazarjani & Payam Piray "Efficient Learning of Predictive Maps for Flexible Planning" (2026) *Nature Communications* PDF

Yi Xu, **Armin Bazarjani**, Hyung-gun Chi, Chiho Choi, Yun Fu "Uncovering the Missing Pattern: Unified Framework Towards Trajectory Imputation and Prediction" (2023) *Conference on Computer Vision and Pattern Recognition (CVPR)* PDF

Experience

Graduate Researcher — June 2023 – Present

University of Southern California, Supervisor: Dr. Payam Piray — Los Angeles, CA

- Extended the default representation, a predictive-representation RL model, to a fully feature-based formulation that preserves compositionality; proved convergence and showed improved task transfer over baselines (*NeurIPS*, 2026).
- Developed an RL model of human decision-making that learns policy-independent predictive maps of the environment online, enabling flexible transfer across tasks. The model gives a single normative account of behavioral effects previously attributed to separate mechanisms and outperforms existing models in formal model comparison on human and rodent behavioral data (*Nature Communications*, 2026).
- Proposed a theory of hippocampal replay as repair of the cognitive map that unifies a range of replay phenomena under a single mechanism; derived a novel prediction from the theory and confirmed it in neuronal replay recordings (manuscript in preparation).
- Augmented EDELINE, a state-of-the-art diffusion-based world model, with an episodic memory buffer, showing that a simple memory component improves agent performance in environments that require long-term memory.
- Used deep RL agents (model-based and model-free) as encoding models of brain activity, comparing how well each agent's internal representations predict neural responses across brain regions.

Research Engineer — Dec. 2021 – Mar. 2023

Honda Research Institute, Supervisor: Dr. Behrzad Dariush — San Jose, CA

- Co-developed the first unified framework to jointly address trajectory prediction and imputation, introducing the benchmark and baselines for the problem (*CVPR*, 2023).
- Developed models for road-scene classification and segmentation in an orthographic bird's-eye-view (BEV) frame, exploring temporal architectures that use long- and short-term context to handle occlusions.
- Fine-tuned and deployed a real-time trajectory prediction model on vehicle hardware, optimizing the pre- and post-processing pipeline for latency and adapting inference to hardware constraints.

Patents

Yi Xu, **Armin Bazarjani**, Hyung-gun Chi, Chiho Choi, Yun Fu "Trajectory imputation and prediction" *Google Patents*, Patent No. US20240160812A1

Earlier Research & Internships

Center for Artificial Intelligence in Society (CAIS), USC Feb – Jun 2021
Undergraduate Researcher, Supervisor: Dr. Bistra Dilkina — Unsupervised clustering and classification of camera-trap images with a GMM-parameterized network; benchmarked against DEC and DeepCluster.

Neuro Imaging with Deep Learning Lab (NIDL), USC Sep 2019 – Jul 2020
Undergraduate Researcher, Supervisor: Dr. Hosung Kim — GAN-based brain aging on structural MRI to model neurodegeneration; improved U-Net and Capsule Net segmentation of white matter hyperintensities with multi-scale foreground highlighting.

Qualcomm Summers 2017, 2018
Software Engineering Intern, Supervisor: Dr. Suhail Jalil — Built a C simulator for facial-recognition hardware and ported existing code to C++. Classified users as adult or child from capacitive touchscreen signals; presented at the Qualcomm ML Summit.

Reviewing

Journals: Nature Communications

Conferences: CVPR

Technical Skills

Languages: Python, C/C++, MATLAB
Frameworks: PyTorch, JAX, TensorFlow
Libraries: NumPy, Pandas, Scikit-learn, Gymnasium, OpenCV, Matplotlib
Developer Tools: Git, Docker